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ABSTRACT. A simple-to-use fluorescent stain, 4′,6-diamidino-2-phenylindole (DAPI), visualizes nuclear DNA in both living and fixed cells. DAPI staining was used to determine the number of nuclei and to assess gross cell morphology. Following light microscopic analyses, the stained cells were processed for electron microscopy. Cells stained with DAPI showed no ultrastructural changes compared to the appearance of cells not stained with DAPI. DAPI staining allows multiple use of cells eliminating the need for duplicate samples.

Key words: 4′,6-diamidino-2-phenylindole, DAPI, nuclear staining, vital stain, cell culture

The polycationic molecule, 4′,6-diamidino-2-phenylindole (DAPI), originally synthesized as a trypanocide (Dann et al. 1971), binds strongly to adenosine-thymidine (A-T) rich regions of nuclear DNA via electrostatic interactions (Kania and Fanning 1976, Hajduk 1976, Kapuscinski and Skoczylas 1978, Kapuscinski and Szer 1979, Tijssen et al. 1982, Manzini et al. 1983, Kapuscinski 1990). DAPI, when bound to nuclear DNA, fluoresces due to the inhibition of internal rotation in DAPI's molecular structure (Kapuscinski and Skoczylas 1977).

DAPI has been used to stain both fixed and live cells. DAPI staining is compatible with aldehyde and ethanol fixation. In confocal microscopy, DAPI is an excellent fluorescent tag as photo-bleaching is insignificant, thus allowing prolonged scanning (Wright et al. 1989). As a vital stain, DAPI is relatively nontoxic and has been used in intraocular injections (Brown and Hitchcock 1989) and in retrograde transport (Yezierski and Bowker 1981, Hancock 1982). DAPI has been used in double labeling experiments because it is compatible with other chromogens such as rhodamine (Tapscott et al. 1989), fluorescein isothiocyanate (Hoff 1988), horseradish peroxidase (Hancock 1982, Yezierski and Bowker 1981), hematoporphyrin (Takahama and Kagaya 1988) and distamycin A (Macera et al. 1989, Lin et al. 1990).

In this report, we present results of studies which utilized DAPI to facilitate counting nuclei and assessing the gross alignment of fixed cochlear sensory hair cells. These DAPI stained cochlear cells were ultimately used for electron microscopic studies. We also used DAPI vital staining to quantify the number of nuclei in dissociated myocytes and cultures of kidney cells.

MATERIALS AND METHODS

Fixed cells

Cochlear surface preparations. Inner ears were excised from Mongolian gerbils ranging from 3–12 months of age. Gerbils were anesthetized with urethane (1.5 g/kg, IP) and were perfused intracardially with 0.1% sodium nitrite in 120 mM phosphate buffer (pH 7.2, 37 C) followed by a fixative consisting of 4% paraformaldehyde and 2% glutaraldehyde in 120 mM phosphate buffer (pH 7.2) at room temper-

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ature. Cochleas were further perfused with the same fixative and dissected for surface preparations according to the method of Schulte and Adams (1989).

Live cells

Kidney cells. Cultures of Wilms' tumor kidney cells were grown as previously described (Garvin et al. 1987).

Myocytes. Swine myocyte cultures were prepared using the procedure of Spinale et al. (1991). Briefly, swine (20–25 kg) were intubated and anesthetized with Isofluorane (1.5%, 1.5 O_2 l/min). A sternotomy was performed, the heart extirpated, and a section of the left ventricular free wall was isolated. The first large obtuse marginal branch of the circumflex coronary artery was cannulated. Using the method of Hewett et al. (1983), the ventricular section (5 × 5 cm) was perfused with an oxygenated Krebs' solution (37 C, pH 7.5) for 10 min followed by a collagenase solution for 35 min. The tissue was then minced into small cubes and added to a fresh Krebs' solution containing DNase and collagenase. Following a 15 min agitation of the tissue suspension, the supernatant was removed and filtered into centrifuge tubes. The cell suspension was allowed to spin for 10 min and the resulting pellet was resuspended in a tissue culture medium (Nutrient Mixture F-12, Gibco Laboratories, Grand Island, NY). Matrigel coated (Collaborative Research Inc., Bedford, MA) coverslips were plated at a concentration of 5×10^5 cells/ml and incubated for 1 hr at 37 C. This procedure yielded > 60% of rod-shaped myocytes for study.

Stock DAPI solution. Ten milligrams of DAPI (Sigma, St. Louis, MO) in 100 ml phosphate buffered saline (PBS) was stirred overnight at room temperature. The mixture was passed through a 0.45 μm nylon filter unit (MSI, Westboro, MA) to remove any insoluble materials. For vital staining, DAPI stock solution was irradiated for 10 min in a gammator (1104 cGy per min) Isomedix, Inc., Parsippany, NJ).

Fixed cell staining. Cells, either aldehyde fixed or fixed with 70% ethanol (fixation time, 5 min), were equilibrated in a minimum amount of PBS. For staining, two or three drops of DAPI stock solution (approximately 10–20 μg/ml) were added to the PBS. Cells were used for fluorescent microscopy without rinsing.

Vital staining. Sterile DAPI stock solution was added to the culture medium at a final concentration of 10 μg/ml. Optimal vital staining required incubation for several hours. In some cases, cultures of kidney cells were maintained with 10 μg/ml DAPI in the medium for a week without any noticeable deleterious effects.

Fluorescence microscopy. Fluorescent microscopic observations were made with a Zeiss IM 35 research microscope equipped with 50 W mercury arc lamp for epifluorescence. The DAPI/DNA complexes were excited with a Zeiss 02 filter set (365 nm light) and the 420 nm epifluorescent emission was viewed and photographed. Fluorescent photographs were taken with Kodak T max 400 film (ASA 400) and transmitted light micrographs were made with Kodak T max 100 film (ASA 100).

Electron microscopy. Both DAPI stained and DAPI free cells were processed for electron microscopy. Following primary fixation, cells were routinely postfixed in osmium (1% OsO4 in PBS, 2 hr, 4 C). In some experiments, the osmium postfixation step was omitted. Fixed cells were rinsed, dehydrated in an ascending series of ethanols, and embedded in Epon resin. Thin sections were stained with uranyl acetate/lead citrate and observed in a JEOL 100S electron microscope.

RESULTS AND DISCUSSION

DAPI stained sensory cells in cochlear surface preparations revealed regular rows of round fluorescent nuclei (Fig. 1A). The orderly rows of nuclei provide a suitable input for automatic segmentation and counting as well as input for modified nearest neighbor measurements for missing cell search. The transmitted light image (Fig. 1B) illustrates the difficulty encountered in delineating individual cells compared to the easily discernible cells viewed under fluorescence (Fig. 1A).

DAPI staining of cell nuclei has significantly reduced the microscope time required for counting and for evaluating gross cell morphology. In hearing studies, sensory cells showing abnormalities, misalignment, or those which are missing often correlate with pathologies induced by noise exposure, ototoxic drugs (i.e., aminoglycoside antibiotics), some disease processes, or age-related hearing losses (presbyacusis). The time required to count approximately 4,000 hair cells per cochlea and to analyze the resulting data averaged

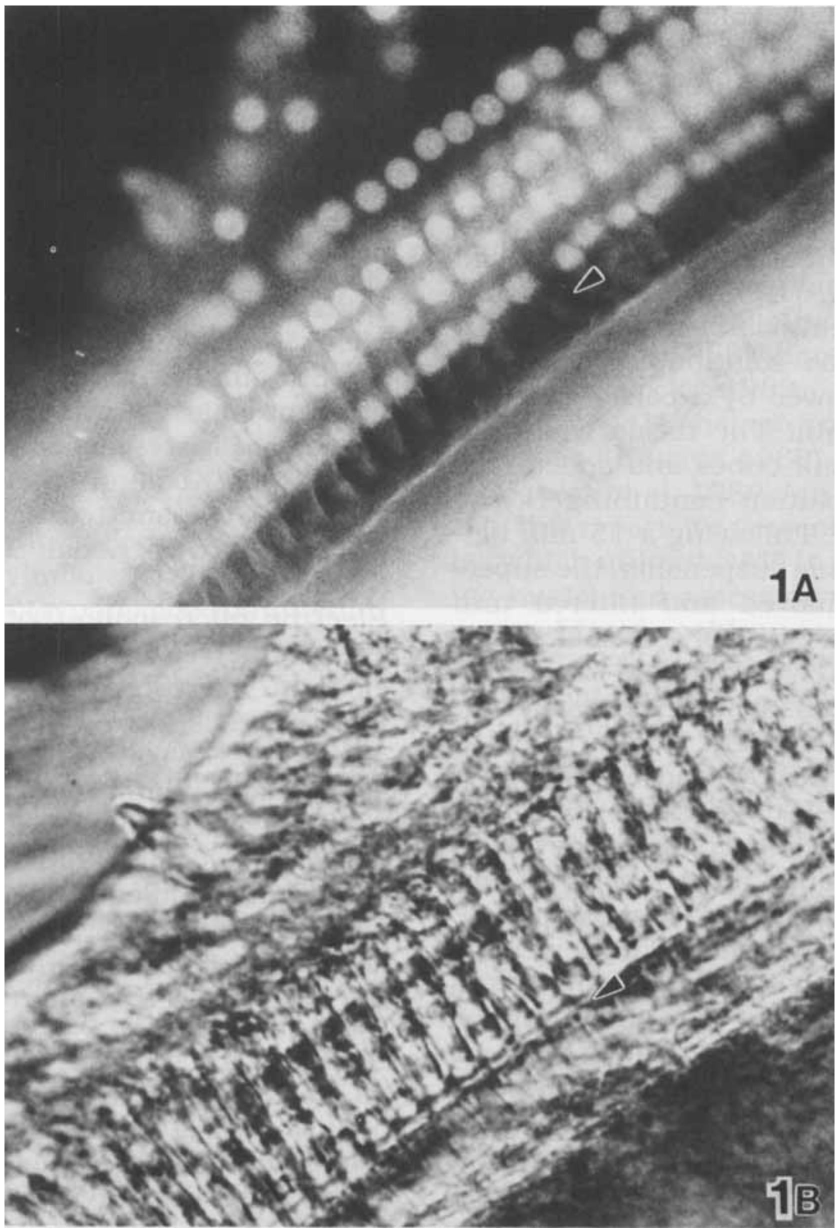

Fig. 1A and B. Surface preparation of a gerbil cochlea used for counting nuclei and assessing gross morphology of hair cells. Arrowhead indicates alteration in cell alignment. (A) Nuclear fluorescence of DAPI stained tissue. The rows of the sensory cell nuclei are easily discernible for quantitative and morphological evaluation. × 90. (B) Transmitted light photomicrograph of DAPI stained tissue (same field as in A). Individual cells are not as easily seen as compared with those in Fig. 1A above. × 90.

about one week using unstained material. This time has been reduced at least by 50%.

We found that DAPI staining does not alter ultrastructure and we were able to use each sample for light and electron microscopy. We processed both DAPI free and DAPI stained cochlear surface preparations for electron microscopy. The ultrastructure of the nuclei as well as mitochondria and plasma membranes in both DAPI free (Fig. 2A) and DAPI stained cells (Fig. 2B) was similar. We concluded that DAPI staining did not label any ultrastructural features such as the nuclear staining noted with Feulgen's stain (Knight and Lewis 1977).

A question arose, however. When DAPI stained material is processed for electron microscopy, does the DAPI stain remain bound to nuclear DNA? To answer that question, we examined thick sections of DAPI stained material for fluorescence. The thick sections showed only a low level fluorescence with no brightly fluorescent nuclei. The low level fluorescence was determined to be background fluorescence,

Table 1. Morphometric Analysis of DAPI Stained Normal Myocytes*

Length (μm)	114.1 ± 2.5**
Width (μm)	27.6 ± 0.5
Number nuclei/myocyte	3.8 ± 0.13

* 150 myocytes total; 50 myocytes from three control animals
** Mean and standard error of the mean

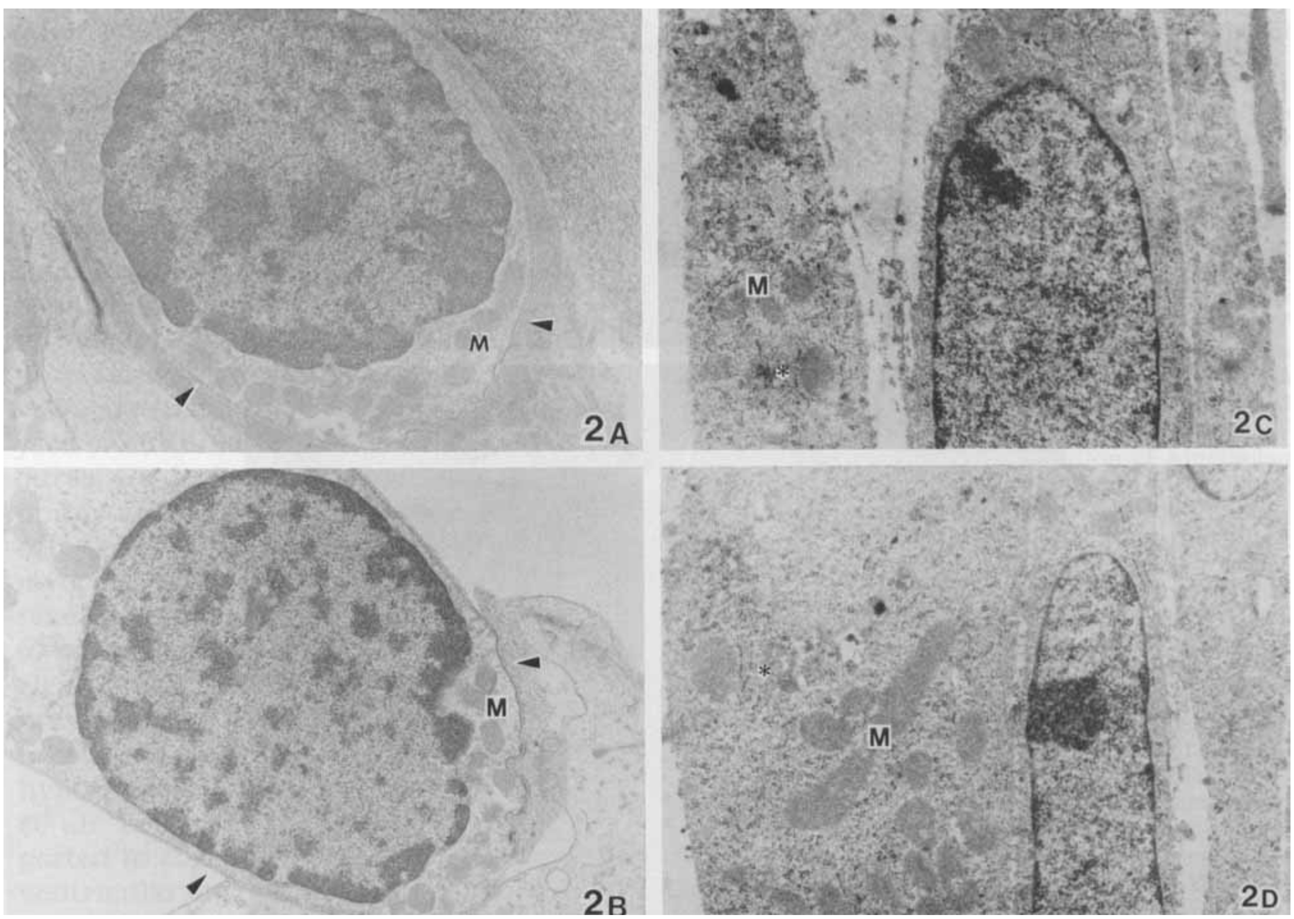

Fig. 2A–D. Electron micrographs comparing DAPI free and DAPI stained cells, M, mitochondria; arrowhead, plasma membrane; asterisk, rough endoplasmic reticulum (RER) (A) DAPI free cochlear sensory cell. The ultrastructure of DAPI free cells appears similar to that of DAPI stained cells (compare Fig. 2(A) and B). × 14,000. (B) DAPI stained cochlear sensory cell. The nucleus, mitochondria, and plasma membrane appear similar to those structures seen in Fig. 2A. × 14,000. (C) DAPI free cultured Wilms' tumor cells from a human kidney. Tissue was processed for electron microscopy without osmium postfixation. The ultrastructure of nuclei, mitochondria, and RER are similar in Fig. 2, C and D. × 23,000. (D) DAPI stained cultured Wilms' tumor cells from a human kidney. Legend is same as in Fig. 2C. × 23,000.

attributable to the Epon embedding resin. To explore the loss of nuclear fluorescence, we aldehyde fixed fresh cells, stained the cells with DAPI, and then processed the cells for electron microscopy. At each step in this routine processing procedure, we monitored the DAPI fluorescence. Initially, the nuclei of the cells displayed the typical, brilliant blue fluorescence. Immediately after osmium postfixation, however, we no longer saw any fluorescent nuclei. Next, we DAPI stained aldehyde fixed cells and processed the cells for electron microscopy using a revised procedure. The revised procedure differed from the routine processing only

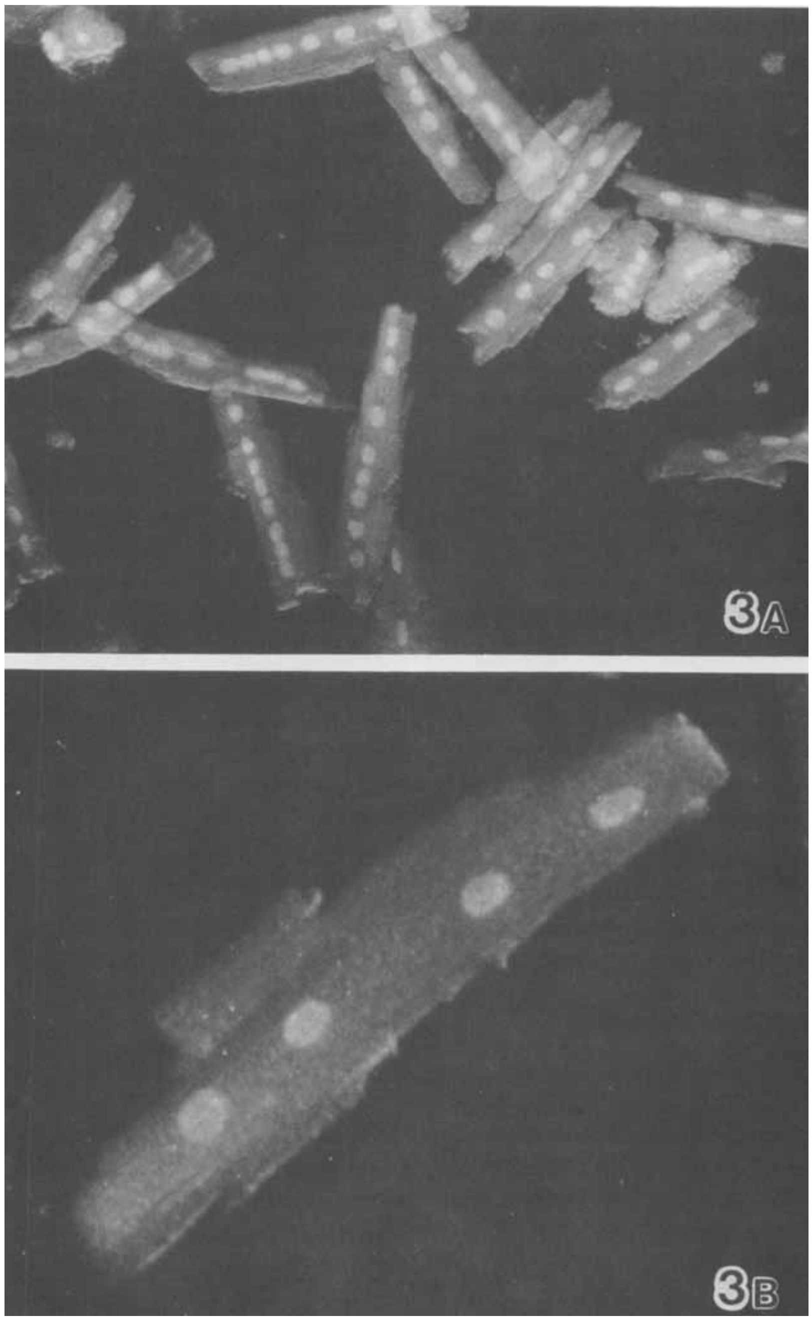

Fig. 3A and B. Cultured pig myocytes stained with DAPI. (A) Light micrograph shows that most myocytes are multinucleated. × 60. (B) An individual myocyte at higher magnification. The average number of nuclei per normal myocyte is four (see Table 1). × 100.

in that the osmium postfixation step was omitted. Thick sections of the nonosmicated, DAPI stained tissue displayed fluorescent nuclei.

We were interested in comparing the ultrastructure of nonosmicated, DAPI free cells with that of nonosmicated, DAPI stained cells. Fig. 2C shows that nonosmicated, DAPI free cells were indistinguishable from nonosmicated, DAPI stained cells (Fig. 2D).

We have not pursued further analysis of the interaction between DAPI and osmium and the associated loss of DAPI fluorescence. It is possible that DNA bound DAPI also binds osmium and that a putative DNA/DAPI/osmium complex has different excitation/emission properties compared to the fluorescent DNA/DAPI complex. Another possible explanation for the loss of DAPI fluorescence in osmicated tissue may involve a competitive binding process between osmic acid and nucleic acid. If DAPI preferentially binds to osmic acid, a resulting osmium/DAPI complex may not fluoresce. Alternatively, such an osmium/ DAPI complex may be fluorescent but, due to solubility in dehydrating solutions, may no longer be present in the tissue following processing.

We also used DAPI in quantitative analyses of cultured myocytes. In these procedures, we determined the interactive minimum and maximum diameter of myocytes and the number of nuclei per myocyte (Table 1). DAPI stained myocytes revealed that normal pig myocytes appear to be multinucleated (Fig. 3A and B). Nuclear proliferation has been documented previously in human and rat myocytes taken from hearts with pressure overload hypertrophy (Astorri et al. 1977, Olivetti et al. 1987). Binucleates have been reported to comprise approximately 90% of ventricular myocytes of rat heart (Rakusan 1984). In our study, we are establishing a database for normal myocytes which will be used to assess possible changes in pathological hearts.

In summary, we find that DAPI is a versatile and simple stain which facilitates quantitative procedures at the light level. Further, since the DAPI molecule does not alter the ultrastructure of cell organelles, DAPI stained cells can be used for ultrastructural studies.

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